Methods Project

Where should you escape in case of a new TB variant

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1 Introduction

1.1. Relevance of the work and data

Tuberculosis (TB) represents a significant health challenge: the World Health Organization has set the goal to reduce its incidence by 90% between 2015 and 2035. Diagnosis, treatment, and prevention of the disease present considerable difficulties, and it remains a leading cause of death globally. According to available data, its endemicity can still be considered under control. Nevertheless, the downward trend, which previously appeared to be leveling off, has shown signs of reversal [8]. In the United States, in 2023, the Centers for Disease Control (CDC) reported an increase of 16% in new cases of TB. This is in addition to increases seen yearly since 2020, after 27 years of decreasing numbers [2]. This project uses probabilistic modeling to explore how structural and health-related variables could influence vulnerability and containment potential in a new TB-like outbreak in different nations.

A wide range of data were used, including epidemiological data from the CDC and WHO, demographic and geographic indicators from the World Bank, and national statistics on chronic diseases and access to healthcare. Mobility data and transportation infrastructure were also considered, along with Bayesian modeling approaches from biomedical research. Together, datasets and research papers offer a comprehensive foundation for analyzing TB vulnerability across different countries.

Understanding how health, demographic and infrastructural factors contribute to a country's vulnerability to epidemics is critical. Traditional modeling methods often fail to handle uncertainty, non-linearity, and missing data, all common in epidemiological settings. Bayesian networks offer a promising solution: they provide a flexible probabilistic framework for modeling causal relationships among multiple variables and are particularly well suited for domains where empirical data may be incomplete or expert knowledge is needed to supplement it [14].

1.2. Modeling Strategy

The project began with the idea of comparing how the impact of the disease might vary between countries, based on the assumption of a spreading variant for which existing vaccines and treatments are ineffective.

We chose to follow the Trigger–Control–Event–Consequence–Mitigant structure and based our study on two country-specific factors that would influence how the impact varies across countries: containment ability and population vulnerability. To support this, we identified key

drivers for each factor, for example, population density (which influences containment ability) and age distribution (which affects vulnerability), using data from reliable sources about TB.

Once these two components were defined, we combined them with the disease's level of contagiousness to develop an index indicating the likelihood of infection. This index represents the core outcome we intended to use for evaluating the potential impact; more specifically the infections and the death toll.

It is important to note that, while the data supporting the individual factors are grounded in established studies on tuberculosis, the formulas for Containment Ability, Population Vulnerability, and Epidemic Index represent core assumptions of our model. This is because, given the hypothetical nature of the new variant, we could not rely on existing studies for those specific dynamics.

2 | Model

In this chapter[1] we present the variables used in our model (2.1), we explain how they are interconnected and what is the reasoning behind them (2.2). Finally, we present the assumptions underlying our model (2.3).

2.1. Nodes Description

Country (C): Labelled node representing the three countries of our study (Italy, USA, China).

Population (P): Integer interval representing the population of each county.

Land Area (LA): Discrete real representing the area of each country.

Population Density (PD): Continuous interval representing the ratio between population and land area of each country.

Age (A): Labelled node representing age distribution of a country's population.

Vulnerability Age (VA): Continuous interval node that combines the probability of a citizen having a specific age with the associated age-related vulnerability to TB.

Travel&Connectivity (TC): Ranked node representing the level of connectivity of each country.

Healthcare Quality and Access Index (HI): Discrete real node representing the quality of the national healthcare system and the access index.

Containment Ability Index (CAI): Continuous interval (between 0–1) modeling a country's epidemic control capacity.

Lockdown (L): Boolean node constituting the state of lockdown of a country.

Chronic Diseases (CD): Boolean node representing the probability that an individual may or may not have a chronic disease.

Vulnerability Chronic Diseases (VC): Boolean node representing the likelihood of high or low vulnerability to TB due to chronic diseases.

HIV (H): Boolean node representing the probability that an individual has HIV.

Vulnerability HIV (VH): Boolean node representing the increased risk of TB due to HIV status.

[1] This chapter presents a summarized description of the nodes and their connections, see Appendix A for a detailed description of the data and formulas we used to map them.

Vulnerability Index (VI): Continuous interval (between 0-1) modeling a country's vulnerability to the disease.

Population Vulnerability (PV): Ranked node representing the level of vulnerability for each population.

Epidemic Index (EI): Continuous interval (between 0-1) which represents the probability of being infected in a country.

Infected (I): Continuous interval indicating the population that was infected in a country.

Contagiousness (CO): Ranked node representing TB contagiousness.

Cure Developed (CD): Boolean node indicating whether a cure was developed in a country.

Cure Efficacy (CE): Ranked node indicating that indicates how effective the cure is.

Death Rate (DR): Discrete real indicating the probability of dying having contracted the disease.

Death Toll (DT): Continuous interval that indicates the number of deaths.

2.2. Model's structure

Model's core

The core of our bayesian network, as shown in the schematic representation in Figure 2.1, consist of the assumed **contagiousness** of the new TB variant (our *trigger*), the **epidemic index** along with the number of **infected** people (our *risk event*), the **containment ability** and **population vulnerability** of the country taken into consideration (our *control* variables), and finally the total **deaths** (the *consequence*) and the nodes related to a possible **cure** (*mitigating* factor).

As we said, we developed our work with the realisation that different countries have different characteristics that make the diffusion of a disease different in each of them. That's why another important node in our network is the **country** node, that changes the country-specific inputs accordingly.

The full bayesian network is shown in Figure 2.2.

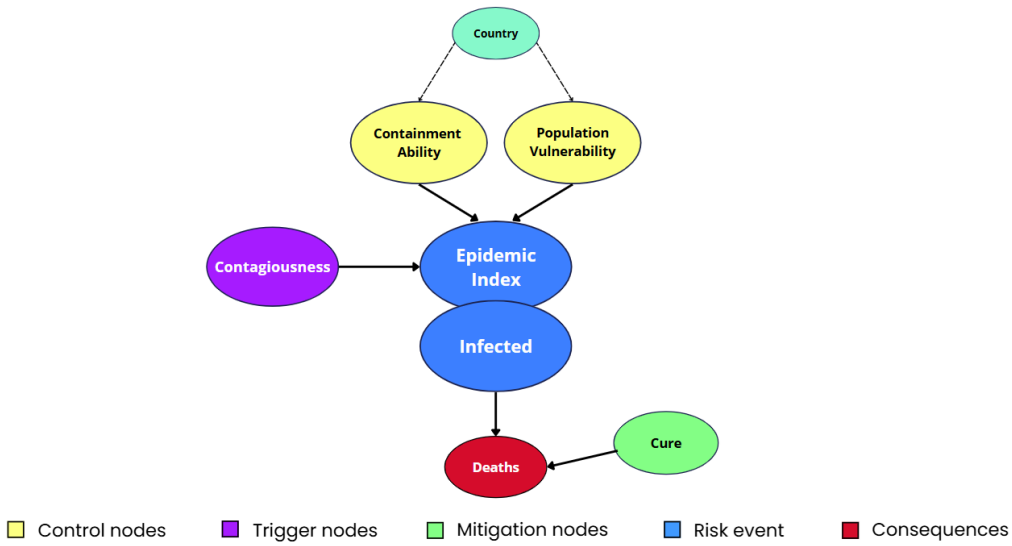


Figure 2.1: Schematic Representation

Control variables

The two main control variables, ***Containment Ability Index*** and ***Population Vulnerability***, are defined by a series of other nodes.

In particular, we modeled the concept of ***Containment Ability*** as dependent on the *Population Density* (*Population* over *Area*), healthcare system conditions, for which we used the well defined standard *Health Quality and Access Index* by the GBD (Global Burden of Diseases), and the level of *Travel and Connectivity* within a country. We included in this part of the network the possibility for a country to mandate a national *Lockdown*, which would causes the connectivity node to fall to "low" and also directly influences the containment ability.

The ***Vulnerability Index***, instead, is comprised of the different vulnerabilities related to three influencing factors in an individual, that makes him more (or less) prone to catching the disease: *Age*, having contracted *HIV*, and the presence of *Chronic Disease* different from HIV. Each of these factors feeds into a corresponding vulnerability node: *Vulnerability Age*, *Vulnerability HIV*, and *Vulnerability Chronic Disease*, which are defined in their Node Probability Tables (NPTs) as the conditional probability of infection given the specific state of the associated factor. The absolute (unconditional) probability of these three nodes depends on the preceding nodes that define the distributions of these vulnerability factors among the population of each country. Finally, the *Vulnerability Index* is converted into a ranked scale ("low", "medium", "high") through the node ***Population Vulnerability*** by dividing the space of its values.

Risk Event and Consequences

The *Epidemic Index* combines the previously defined control nodes with the *Contagiousness* node and represents the probability of an individual contracting the infection. By multiplying the *Epidemic Index* by the *Population* we obtain the estimated number of infections, our core risk event. Finally, the death toll is calculated as the product of the number of infections and the *Death Rate*.

Cure nodes (Mitigation)

As mitigation variables we modeled the possibility of development of a cure for the disease, through the boolean *Cure Development* node. To make the model more realistic we included a ranked node also to describe the *Cure Efficacy*, which influences the final *Death Toll*, by directly affecting the *Death Rate* of the infected people.

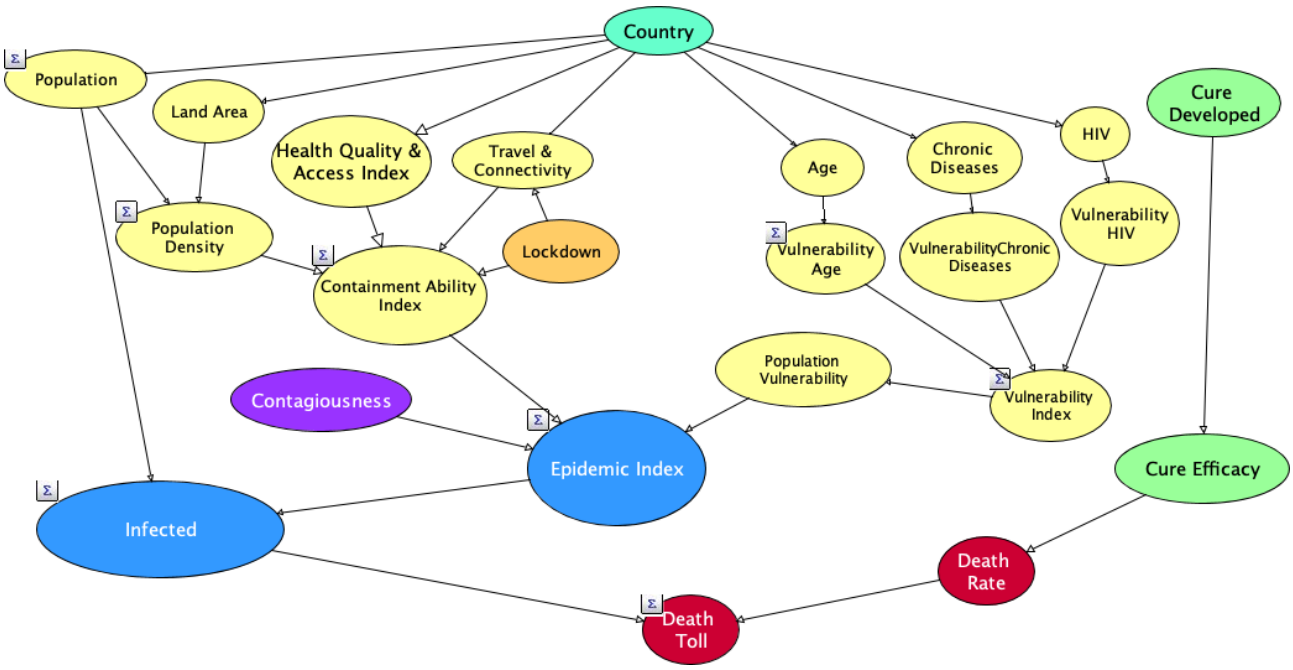


Figure 2.2: Bayesian Network

2.3. Assumptions

1. No vaccination is assumed. The model reflects a novel, resistant TB variant, where existing treatments or vaccines are ineffective.
2. Patient zero is equally likely to be in each country. We assume no bias in the initial outbreak location. This is modeled with equal prior probabilities across all three countries.
3. *CAI* is calculated by treating all its input variables as equally impactful. The formula incorporates *Healthcare Index*, *Population Density*, *Travel&Connectivity*, and *Lockdown*

with equal structural weight. However, in reality, some variables (e.g. *Population Density*) might play a more dominant role. Moreover, the numerical values assigned to qualitative variables in *Travel&Connectivity* were defined subjectively by the team.

4. *Lockdown* effectiveness is standardized, increases the *Containment Ability Index (CAI)* by a fixed multiplier (1.2) when active. This simplifies the heterogeneity in policy stringency and public compliance across nations.

5. *Contagiousness*, *Containment Ability Index*, and *Vulnerability Index* are assumed to have equal weight in determining the *Epidemic Index*. This is a modeling assumption, there is no empirical evidence suggesting that all three factors contribute equally.

6. The nodes *Cure Developed* and *Lockdown* should ideally be linked to the *Country* node, but in our model they are treated as standalone elements.

7. *Contagiousness* is modeled as independent from *Cure Efficacy*. An effective treatment could reduce the contagiousness of the disease. However, this potential interaction is not captured in our model, where contagiousness is treated as an intrinsic, fixed property of the disease.

3 | Weaknesses

1. To reduce the number of nodes we used a limited number of risk factors compared to the ones analyzed in available literature like Delgado et al. [16] or Narasimhan et al.[5].

2. In our model, we simplified by omitting certain causal relationships. For example, while age is a strong determinant of chronic diseases, we did not include a direct dependency between these nodes. This may lead to confounding bias or misinterpretation of observed correlations, similar to Simpson's Paradox.

3. The previous point is particularly true for the nodes *Contagiousness* and *Death Rate*, which are designed as an intrinsic characteristic of the disease although they are usually dependent on many social, personal, and environmental factors. *Death Rate* would also highly affect *Containment Ability*, *Contagiousness*, and therefore the number of infections.

4. Concerning the *Travel&Connectivity* node, the current model assesses internal connectivity but overlooks international mobility like air travel, which can accelerate global disease spread. The proxy used to estimate the node only addresses the total road network length of a country and its total area. Future versions should include measures like air traffic volume or global mobility to better capture epidemic risks.

5. The model does not include a time component.The outcomes are computed as aggregated values without temporal evolution. As a result, it cannot represent dynamic phenomena such as peak infection times, healthcare system overload, or the delayed effects of interventions.

6. For nodes regarding TB itself, the academic literature often presents really narrow studies with small samples or really specific. For example, the node *Vulnerability HIV* is based on a study carried out in Tanzania. This introduces results that may be study-specific and not generalizable.

4 | Scenarios

In this chapter we present some scenarios and leverage our model to solve them. For clarity, we present only the answers of the scenarios here, the full results of the model can be found in Appendix B.

Scenario 1: "The Silent Variant"

A new variant of tuberculosis emerges exhibiting the same characteristics as traditional TB. Unfortunately, the available vaccines do not work and conventional antibiotics are ineffective. This disease presents the same mortality rate as the current value for standard untreated TB: (50%).

Node Cases Changes:

- Cure development: False
- Lockdown: False
- Country: We run the model each time with a different country

Q1: ***In which country would the disease have more infected people with respect to the population?***

Italy: 1 200 900 infected (*stdev:639 800*) = **2%**
USA: 4 417 200 infected (*stdev:2 440 700*) = **1.33%**
China: 23 594 000 infected (*stdev:12 690 000*) = **1.67%**

ANSWER: Italy

Q2: ***Which one would suffer the highest number of deaths?***

Italy: **600 570** (*stdev:320 550*)
USA: **2 210 400** (*stdev:1 225 100*)
China: **11 804 000** (*stdev:6 368 800*)

ANSWER: China

Considerations: The first outcome is heavily influenced by the *Containment Ability* factor. According to the simulation results, Italy has weaker containment capabilities compared to other countries, while *Population Vulnerability* remains fairly consistent across nations. The second result is more straightforward, as China's higher number of fatalities is primarily due to its significantly larger population.

Scenario 2: "A Secret Cure vs a Strict Lockdown"

Just weeks after the outbreak of the drug-resistant TB variant, the United States government decided to pursue a cure independently and, surprisingly, succeeded in developing a highly effective treatment. Following the U.S. withdrawal from the WHO in 2025 and rising geopolitical tensions over trade, the administration chose not to share the cure with other countries. In response, Italy, fearing the disease's consequences, implemented a strict lockdown.

Node Cases Changes:

ANALYSIS 1:

- Country: USA
- Cure development: True
- Cure Efficacy: High
- Lockdown: False

ANALYSIS 2:

- Country: Italy
- Cure development: False
- Lockdown: True

Q1: ***How much the death toll in USA going to change?***

With this scenario, the estimated number of deaths in USA is 884 090, with a **decrease** of **almost 60%**

Q2: ***Was the lockdown in Italy more effective than the cure developed in USA at reducing the death toll?***

The death toll in Italy with the restriction is 111 500 representing a decrease of 81%. Therefore the **lockdown in Italy is more effective** than the development of a high effective cure in USA.

Considerations: The model suggests that the reason Italy's lockdown proves more effective than the US cure is likely due to the containment issues identified in the previous scenario. The lockdown significantly reduces Italy's epidemic index, which in turn lowers the number of infections. This reduction in cases ultimately has a greater impact on the death toll than simply decreasing the fatality rate while infections remain high, as seen in the US.

Scenario 3: "Aftermath Analysis"

After the virus had subsided and no new cases were reported, the Chinese government wants to issue a report to analyze the impact of the outbreak and the decisions made. Although a minimally effective treatment was developed during the epidemic, the government chose not to implement a lockdown due to economic concerns. This decision ultimately led to losses amounting to 50 million. During the outbreak, they also faced significant challenges in accurately tracking all infections and are now seeking a plausible estimate of the true number of cases

Cases:

- Country: China
- Cure development: True
- Cure Efficacy: Low
- Lockdown: False
- Death Toll: 50 000 000

Q1: ***Given this data, what is a plausible estimate of the infections that occurred during the crisis?***

The model predicts **124 690 000** millions infected with a standard deviation of 681 040.

Q1: ***Can you make considerations on the variant of the disease?***

The belief update suggests that in order to reach this amount of deaths the variant in China must have been **highly contagious**, easier transmissible than the standard strain. Furthermore, the results show that the Chinese population was more vulnerable to the disease than the other countries.

Considerations: The results obtained by entering evidence into the death toll node are quite insightful and offer a deeper understanding of the epidemic. As mentioned earlier, the information propagation triggered by the evidence, updates not only the probability of *Contagiousness* in China, but also *Population Vulnerability* showcasing why Bayesian Networks are powerful tools for this kind of analysis.

Scenario 4: "Worst and Best Scenarios for Italy"

We want to analyze the aftermath, in terms of number of deaths, of the two most extreme possible outcomes for Italy.

Worst Case:

- Country: Italy
- Contagiousness: High
- Cure development: False
- Lockdown: False

Best Case:

- Country: Italy
- Contagiousness: Low
- Cure development: True
- Cure Efficacy: Very High
- Lockdown: True

Worst case 858 190 deaths, best case 52 603 deaths with a **98.8% decrease**.

Considerations: This scenario highlights the flexibility of our model and demonstrates how entering new evidence can significantly impact the final outcomes.

5 | Conclusion

This work aimed to explore how structural, health, and behavioral variables influence a country's ability to respond to a new TB variant outbreak. By leveraging Bayesian networks, we created a probabilistic model that can simulate various epidemiological outcomes based on diverse national contexts and intervention strategies.

From the analysis of the four defined scenarios, the following key considerations are derived:

- High variability in simulation outcomes. Across all scenarios, the model exhibits very high standard deviations in the number of infections and deaths. This reflects the intrinsic uncertainty encoded in our probabilistic framework, and confirms the relevance of scenario analysis for understanding the possible range of outcomes.
- Death toll is strongly influenced by mid-layer variables. While we constructed a complex network of interdependent nodes, we observed that the final outputs, especially the number of deaths, are more directly shaped by the *Epidemic Index* and *Cure Efficacy* than by upstream demographic or health parameters. This suggests that even small variations in intermediate indices can significantly alter the model's forecasts.
- Limited influence of control variables, but wide scenario divergence. Although the *Containment Ability Index (CAI)* does not vary meaningfully across countries, the model proves effective in generating sharply divergent outcomes across scenarios, demonstrating its robustness and capacity to simulate real-world epidemic complexity.

The model, while simplified in structure and reliant on several assumptions, succeeds in capturing essential dynamics of epidemic spread and response. It provides a valuable sandbox for comparative scenario evaluation, and offers useful insights for policymakers regarding the levers that matter most when preparing for high-impact infectious disease events.

A | Appendix A

Country (C):

- Type: Labeled, three states Italy, China, USA.
- Probability: Each state has 33%.
- Description: Countries of our study.

Population (P):

- Type: Simulation Node, Integer Interval.
- Probability: Partitioned Expression on Country:
  - USA: Normal(*mean*=331449281, *variance*=430884)
  - Italy: Normal*(mean*= 58934000, *variance*=292313)
  - China: Normal(*mean*=1408000000, *variance*=872960)
- Description: Population of each country. The node is defined as a Simulation Node with a normal distribution to reflect real-world demographic variability. The standard deviation was estimated by calculating the difference between the average birth rate and the average death rate over a five-year period.
- Sources: [24]

Land Area (LA):

- Type: Discrete Real with 3 states.
- Probability: Node Probability Table:
  - USA: 9525067
  - Italy: 302110
  - China: 9596960
- Description: Land Area of each country in km^2. The node probability table has the three aforementioned states, each state has probability equal to one for its area and equal zero for the others.
- Sources: [22]

Population Density (PD):

- Type: Simulation Node, Continuous Interval.
- Probability: Arithmetic Expression:
  - Population / Land Area
- Description: The PopulationDensity node represents the ratio of population to land area and is defined as a Simulation Node with an arithmetic expression: Population / Land Area.

Age (Age):

- Type: Labelled.
- 5 intervals:
  - 0-14
  - 15-24
  - 25-44
  - 45-64
  - 65+

Age Group	Italy	China	USA
0–14	0.13	0.16	0.17
15–24	0.10	0.12	0.14
25–44	0.25	0.29	0.26
45–64	0.22	0.29	0.25
65+	0.30	0.14	0.18

Table A.1: Age distribution by country (proportional values)

- [24]

Vulnerability Age (VA):

- Type: Simulation Node, Continuous Interval.
- Probability: Partitioned expression with a truncated (0,1) normal distribution for each age.
- Description: Based on CDC surveillance data, the node reflects the increasing vulnerability to TB as age increases, with older age groups showing significantly higher average rates. The distribution parameters (mean and standard deviation) where estimated from a last-10-years time series extracted from the CDC dataset, in which we reasonably assumed there were no trends.

		0–4			5–14			15–24			25–44			45–64			≥65		
Year	Total cases (*1000,	No.	(%)	Rate	No.	(%)	Rate	No.	(%)	Rate	No.	(%)	Rate	No.	(%)	Rate	No.	(%)	Rate
2023	9.633	231	(2.4)	1.20	235	(2.4)	0.6	1.021	(10.6)	2.3	3.011	(31.3)	3.3	2.605	(27.0)	3.2	2.528	(26.2)	4.3
2022	8.332	199	(2.4)	1.1	163	(2.0)	0.4	844	(10.1)	1.9	2.453	(29.4)	2.7	2.42	(29.0)	2.9	2.252	(27.0)	3.9
2021	7.866	159	(2.0)	0.8	156	(2.0)	0.4	675	(8.6)	1.6	2.263	(28.8)	2.5	2.409	(30.6)	2.9	2.204	(28.0)	3.9
2020	7.17	166	(2.3)	0.9	152	(2.1)	0.4	687	(9.6)	1.6	2.114	(29.5)	2.4	2.174	(30.3)	2.6	1.877	(26.2)	3.4
2019	8.895	215	(2.4)	1.1	150	(1.7)	0.4	851	(9.6)	2	2.611	(29.4)	3	2.657	(29.9)	3.2	2.411	(27.1)	4.5
2018	8.996	184	(2.0)	0.9	188	(2.1)	0.5	869	(9.7)	2	2.723	(30.3)	3.1	2.687	(29.9)	3.2	2.345	(26.1)	4.5
2017	9.066	228	(2.5)	1.1	202	(2.2)	0.5	842	(9.3)	2	2.752	(30.4)	3.2	2.75	(30.3)	3.3	2.29	(25.3)	4.5
2016	9.238	224	(2.4)	1.1	163	(1.8)	0.4	934	(10.1)	2.1	2.827	(30.6)	3.3	2.84	(30.7)	3.4	2.249	(24.3)	4.6
2015	9.539	244	(2.6)	1.2	196	(2.1)	0.5	936	(9.8)	2.1	2.859	(30.0)	3.4	3.021	(31.7)	3.6	2.283	(23.9)	4.8
2014	9.379	263	(2.8)	1.3	195	(2.1)	0.5	959	(10.2)	2.2	2.821	(30.1)	3.4	2.953	(31.5)	3.5	2.188	(23.3)	4.7
2013	9.556	294	(3.1)	1.5	188	(2.0)	0.5	979	(10.2)	2.2	2.964	(31.0)	3.6	2.953	(30.9)	3.6	2.178	(22.8)	4.9
2012	9.908	261	(2.6)	1.3	226	(2.3)	0.5	1.018	(10.3)	2.3	3.117	(31.5)	3.8	3.102	(31.3)	3.7	2.183	(22.0)	5.1
AVG				***1.13***			***0.47***			***2.03***			***3.14***			***3.26***			***4.43***
SD				***0.20***			***0.07***			***0.23***			***0.43***			***0.33***			***0.48***

Figure A.1: Excel Sheet

- Sources: Tubercolosis Age [1].

Travel & Connectivity (TC):

- Type: Ranked, 3 states (low to high).
- Probability: NPT
  - USA: Medium
  - Italy: High
  - China: Medium
- Description: The Travel & Connectivity node assesses how easily people and goods move within and across a country. To estimate it, a proxy is used based on the ratio between the *total road network length* (in kilometers) and the *total area of the country* (in square kilometers).
- Sources: Country Surface [23], Network Length [11], Mobility and Spread of Diseases [7].

Healthcare Quality and Access Index (HI):

- Type: Discrete real.
- Probability: NPT
  - USA: 80
  - Italy: 91
  - China: 68
- Description: The Healthcare Quality and Access Index is modeled as a node with a fixed value for each country.
- Sources: [26]

Containment Ability Index (CAI):

- Type: Simulation Node, Continuos Interval
- Probability: Partitioned Expression on Travel&Connectivity and Lockdown:
  - Formula: round((HI/100.0)*(1.0-min(1.0,PD/1000.0))*TC*L*100.0)/100.0
  - Travel&Connectivity:
    - Low: 1.0
    - Medium: 0.7
    - High: 0.4
  - Lockdown:
    - True: 1.2
    - False: 1.0
- Description: This formula captures the key factors affecting a nation's ability to contain an epidemic:
  - Healthcare Index is normalized (/100) to a 0–1 scale, where higher values improve containment.
  - Population Density is penalized via (1 - min(1, Population Density / 1000)), reducing the index as density increases.
  - Travel&Connectivity is a labelled node mapped to numerical weights.
  - Lockdown is a Boolean node that increases containment when active. If Lockdown is True, a 20% boost (L = 1.2) is applied, reflecting enhanced control during restrictions.

  The resulting value, constrained within the range [0.0, 1.0], is modeled using a T-normal distribution with a small variance (0.005) to account for inherent uncertainty while preserving stability. This allows for a continuous probabilistic representation of containment capacity across simulations.

Lockdown (L):

- Type: Boolean (True or False)
- Description: This node was introduced to influence containment ability in case of a lockdown in a country. The probability does not represent the actual probability of a country being in lockdown: probabilities are just placeholders.

Chronic Diseases (CD):

- Type: Boolean (True or False).

- Probability: NPT
  - USA: T=60%
  - Italy: T=40.9%
  - China: T=45.6%
- Description: Probability of having one or more chronic diseases in a given country.
  True=The individual has one or more chronic diseases,
  False=The individual is free from chronic diseases.
- Sources: USA's Chronic Diseases [3], Italy's Chronic Diseases [12], China's Chronic Diseases [19].

Vulnerability Chronic Diseases (VC):

- Type: Boolean.
- Probability: NPT:

		HIV	
		False	True
Vulnerability	High	0.1	0.28
	Low	0.9	0.72

Table A.2: Node Probability Table for Vulnerability Chronic Diseases

- Description: The node estimates the vulnerability for TB in patients with chronic diseases by averaging the risks of the most common conditions, resulting in a 2.84× higher vulnerability to TB. High vulnerability was estimated by setting a baseline for healthy individuals and scaling it by the 2.84× risk for those with chronic diseases. The focus is on the proportion, not absolute values, to preserve the relative impact in the model.

Diseases	Incidence	AVG incidence	Total AVG
Diabetes mellitus	2–3x	2.5x	2.84x
Autoimmune diseases	4x	4x	
Chronic Kidney Disease	1.6–3.6x	2.6x	
Malignant tumor	1.5–3x	2.25x	

Table A.3: Estimated TB incidence increase by chronic disease

- Sources: [25], [4], [6], [17].

HIV (H):

- Type: Boolean.
- Probability: Partitioned Expression on Country:

  - USA: T=0.36%
  - Italy: T=0.25%
  - China: T=0.046%

- Description: Probability of getting the disease, based on the number of cases divided by the population. True=HIV-Positive, False=HIV-Negative.
- Sources:[10], Italy [21], China [18].

Vulnerability HIV (VH):

- Type: Boolean.
- Probability: NPT:

		HIV	
		False	True
Vulnerability	High	0.03	0.87
	Low	0.97	0.13

Table A.4: Node Probability Table for Vulnerability HIV

- Description: Represents the increased susceptibility of individuals to developing active tuberculosis (TB) as a result of their HIV status. The probability of high vulnerability was estimated by assigning a baseline risk to HIV-negative individuals and scaling it by the known relative risk (29× [13]) for HIV-positive individuals. The absolute values are less important than the proportion between the two, which preserves the relative impact of HIV on TB vulnerability within the model.
- Sources: [13].

Vulnerability Index (VI):

- Type: Simulation Node, Continuous Interval:
- Probability: Partitioned Expression on Vulnerability Chronic Diseases and Vulnerability HIV

		VHIV	
		High	Low
VChronicDiseases	High	VAge + 0.5	VAge + 0.25
	Low	VAge + 0.25	VAge

Table A.5: Node Probability Table for Vulnerability Index

- Description: This node represents an individual's overall vulnerability to developing active TB, combining the effects of age, chronic diseases, and HIV. The base vulnerability is

given by the age-related risk (VAge), and it is adjusted additively based on the presence of additional risk factors:

- If both VC and VH are high, the vulnerability increases significantly (VAge + 0.5).
- If only one of the conditions is high, a smaller adjustment is applied (VAge + 0.25).
- If both are low, vulnerability remains unadjusted (VAge).

The adjustments (0.25 and 0.5) are arbitrarily chosen scaling factors.

Population Vulnerability (PV):

- Type: Labelled
- Probability: Expression
- Formula: if(VI<0.1,"Low",if(VI>0.4,"High","Medium"))
- Description: This node translates the continuous value of the Vulnerability Index (VI) into a categorical label representing the overall vulnerability level of a population segment. The thresholds used for classification (0.1 and 0.4) were set by the team. This classification helps simplify the interpretation of vulnerability across countries and is used as an input for evaluating epidemic impact in the model.

Epidemic Index (EI):

- Type: Simulation node, Continuous Interval.
- Probability: Partitioned Expression on Contagiousness and Population Vulnerability
- Formula: $EI = (1.0 - CAI) * CO * V$

		Contagiousness		
		Low	Medium	High
Population Vulnerability	Low	(1.0-CAI)*0.1*0.1	(1.0-CAI)*0.2*0.1	(1.0-CAI)*0.3*0.1
	Medium	(1.0-CAI)*0.1*0.2	(1.0-CAI)*0.2*0.2	(1.0-CAI)*0.3*0.2
	High	(1.0-CAI)*0.1*0.3	(1.0-CAI)*0.2*0.3	(1.0-CAI)*0.3*0.3

- Description: The Epidemic Index quantifies the potential impact of an outbreak, reflecting the interplay between a country's containment capacity (CAI), the contagiousness of the disease, and the vulnerability of the population. The index increases when:
  - The containment ability is low (i.e., CAI close to 0)
  - The disease is highly contagious
  - The population is more vulnerable (e.g., due to age, comorbidities, or HIV prevalence)

The model assumes multiplicative independence among these variables, and the numerical weights assigned to Contagiousness and Population Vulnerability levels were heuristically defined to reflect relative severity rather than based on statistical regression. This simplification allows for intuitive interpretation and scenario comparison but does not capture complex nonlinear dynamics.

Infected (I):

- Type: Simulation node, Continuous Interval.
- Probability: Partitioned Expression on Population and Epidemic Index
- Formula: P * EI
- Description: Represents the estimated number of infected individuals in the population. It reflects how widely the disease has spread within each country.

Contagiousness (CO):

- Type: Ranked, 3 states (low to high).
- Probability:
  - Low: 0.2
  - Medium: 0.5
  - High: 0.3
- Description: Due to the contagious nature of the disease, we initially considered using the basic reproduction number R_0 as a discrete real value. However, studies have shown that estimating an exact R_0 value for TB can be challenging [15]. Therefore, we chose to represent contagiousness as a ranked variable instead. This approach allows us to account for variations among different strains and to reflect findings from the literature, which suggest a higher likelihood of medium to high contagiousness.

Cure Developed (CD):

- Type: Boolean (True or False).
- Description: This probability does not represent the actual probability of developing a cure, but it is only used when approaching the scenarios. Therefore, probabilities are just placeholders.

Cure Efficacy (CE):

- Type: Ranked, 5 states (Null, Low, Medium, High, Very High).
- Description: If the cure is not developed the efficacy is Null. If it is developed, probability is equally distributed in the cure efficacy being Low, Medium, High or Very High. Again,

those probabilities are just placeholders.

Death Rate (DR):

- Type: Discrete Real.
- Probability: Partitioned expression on cure efficacy:
  - Null: 0.5
  - Low: 0.4
  - Medium: 0.3
  - High: 0.2
  - Very High: 0.1
- Description: In untreated cases, the death rate is approximately 50 [20]. Multidrug-resistant (MDR) TB significantly increases the risk, often exceeding 20% [5]. In contrast, treated TB cases in well-developed countries can have rates as low as 1% [9, 20]. We decided to adopt a value of 0.5 to align with the death rate of the untreated TB and to our scenario of a new variant.

Death Toll (DT):

- Type: Simulation node, Continuous Interval.
- Formula: I * DR
- Description: The node represents the estimated number of fatalities resulting from the outbreak. It is calculated as the product of the total number of infected individuals (I) and the death rate (DR), providing a continuous estimate of mortality under given scenario conditions.

B | Appendix B

B.1. Scenario 1: Results

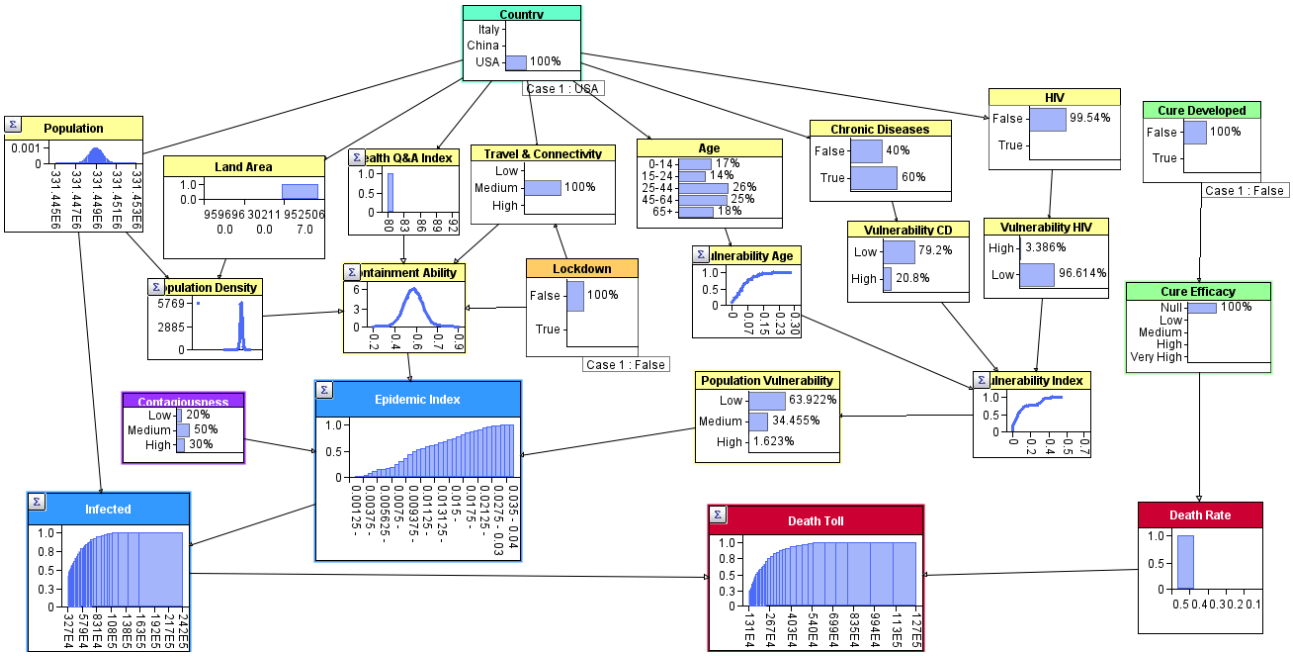


Figure B.1: Scenario 1, Case USA

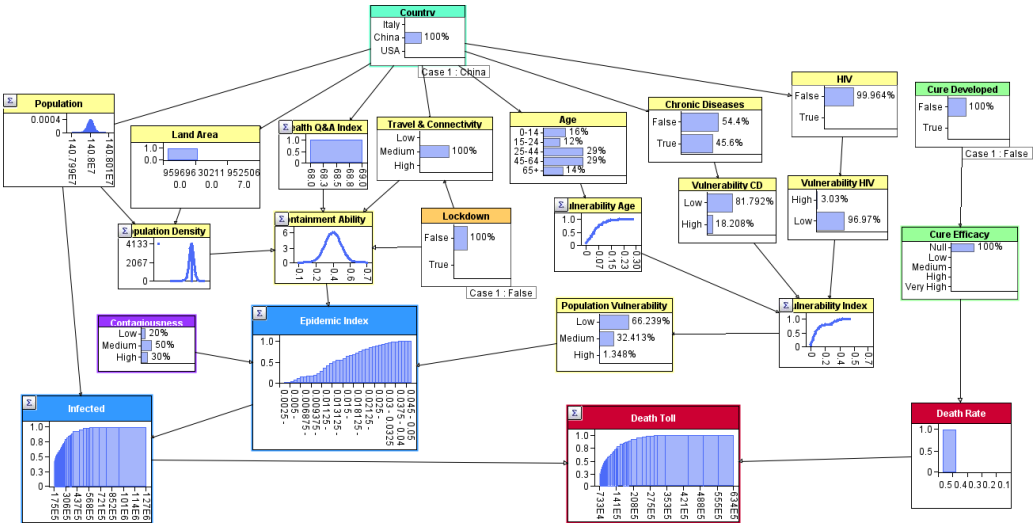


Figure B.2: Scenario 1, Case China

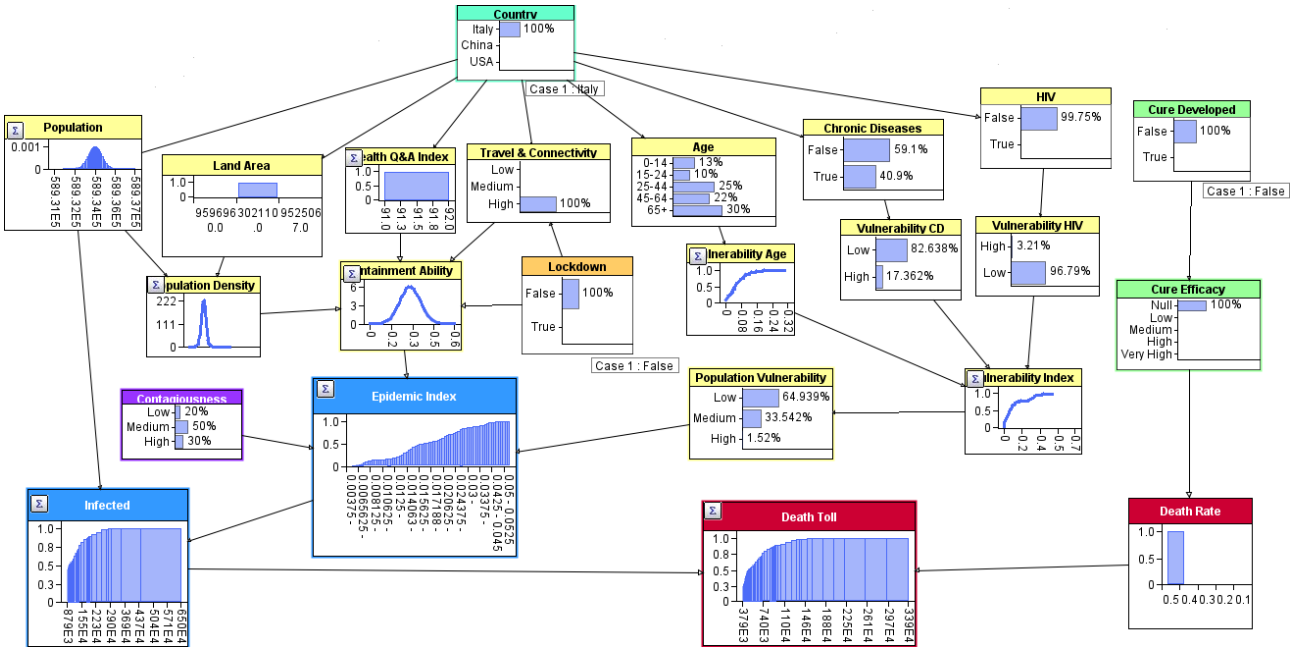
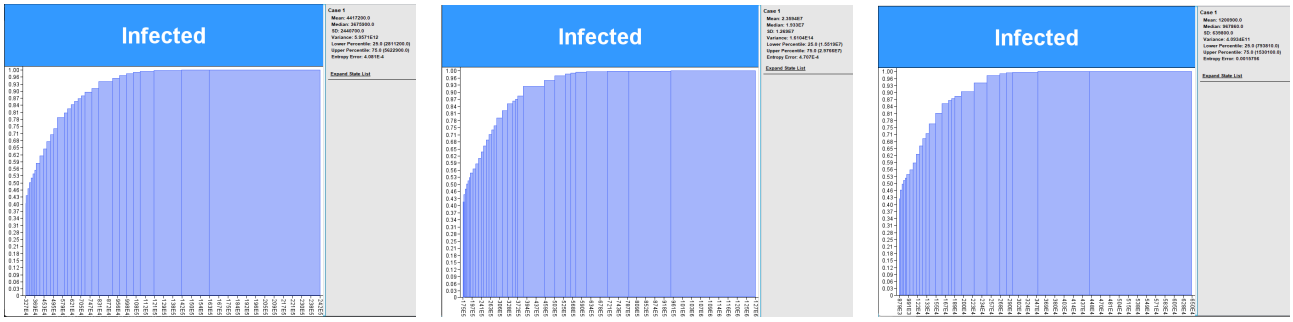
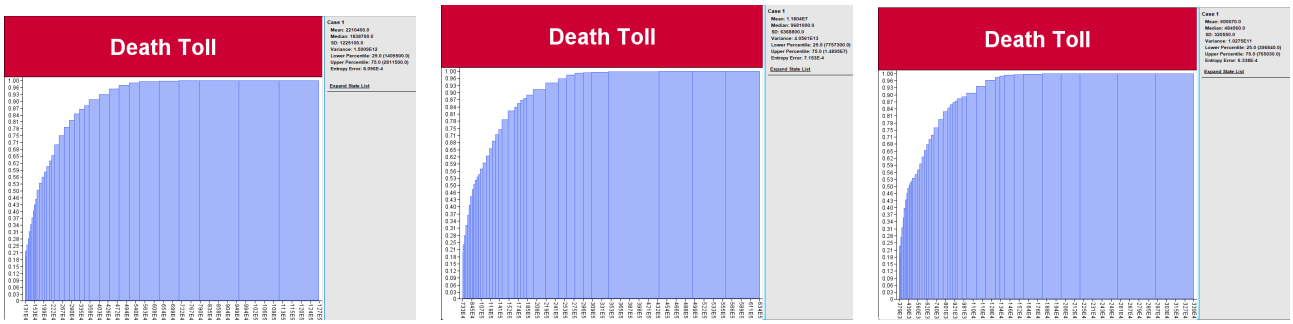


Figure B.3: Scenario 1, Case Italy



(a) USA (b) China (c) Italy

Figure B.4: Scenario 1: Infections



(a) USA (b) China (c) Italy

Figure B.5: Scenario 1: Deaths

B.2. Scenario 2: Results

B.2.1. Question 1

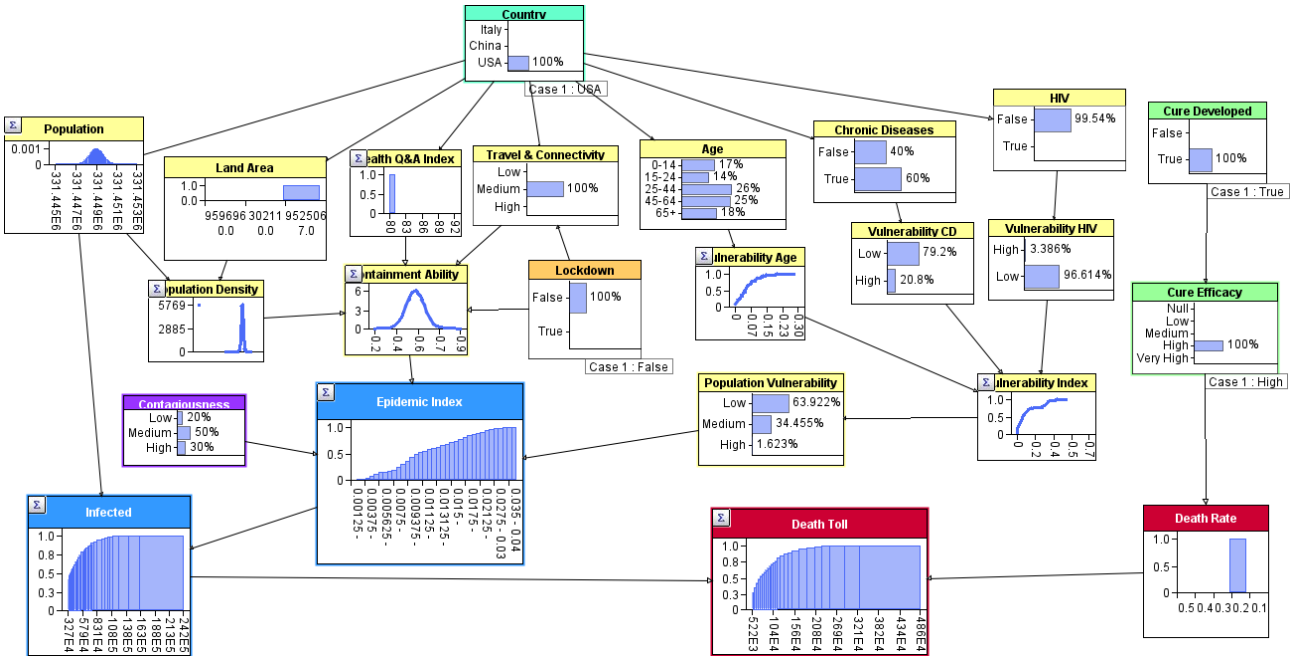


Figure B.6: Scenario 2, Question 1

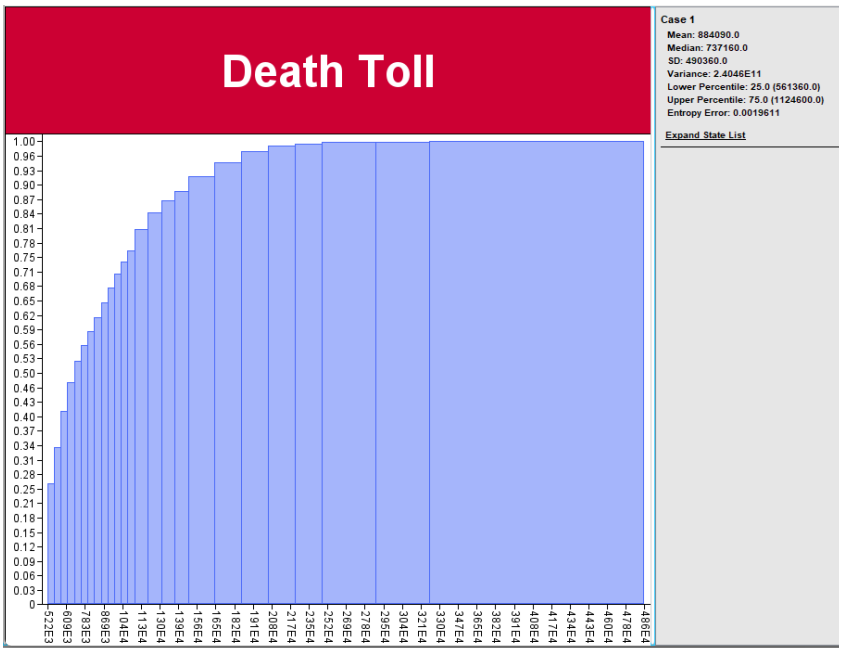


Figure B.7: Scenario 2, Question 1; Deaths

B.2.2. Question 2

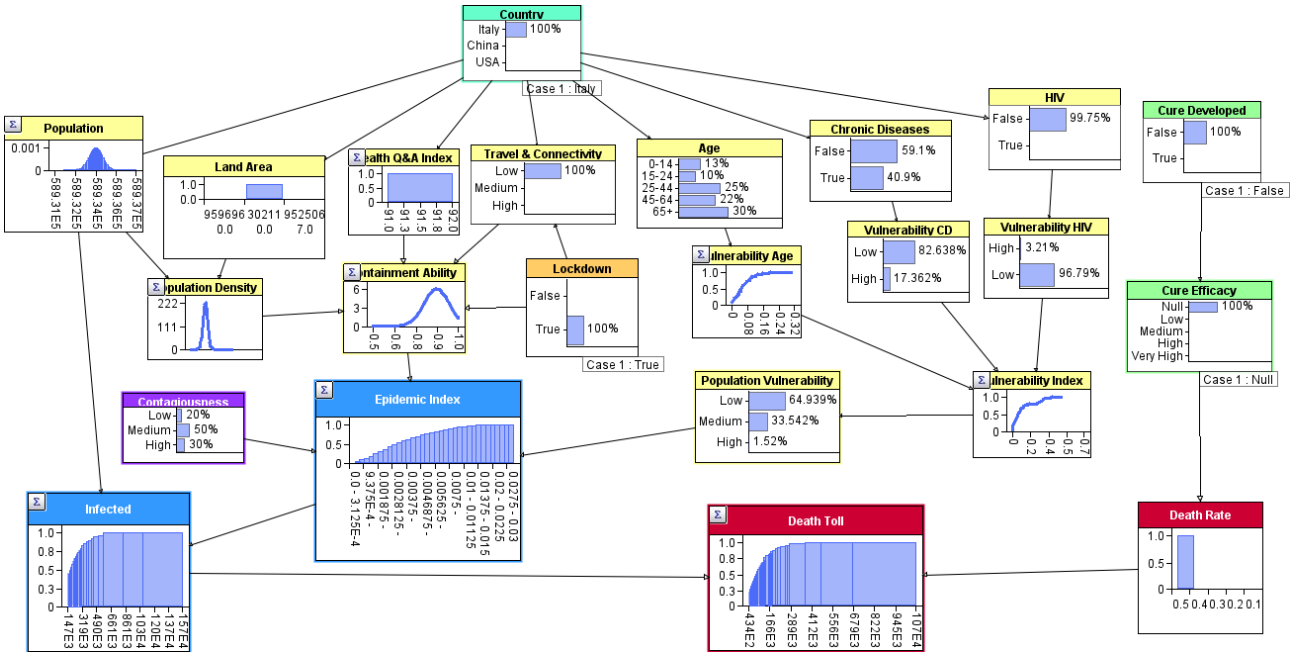


Figure B.8: Scenario 2, Question 2

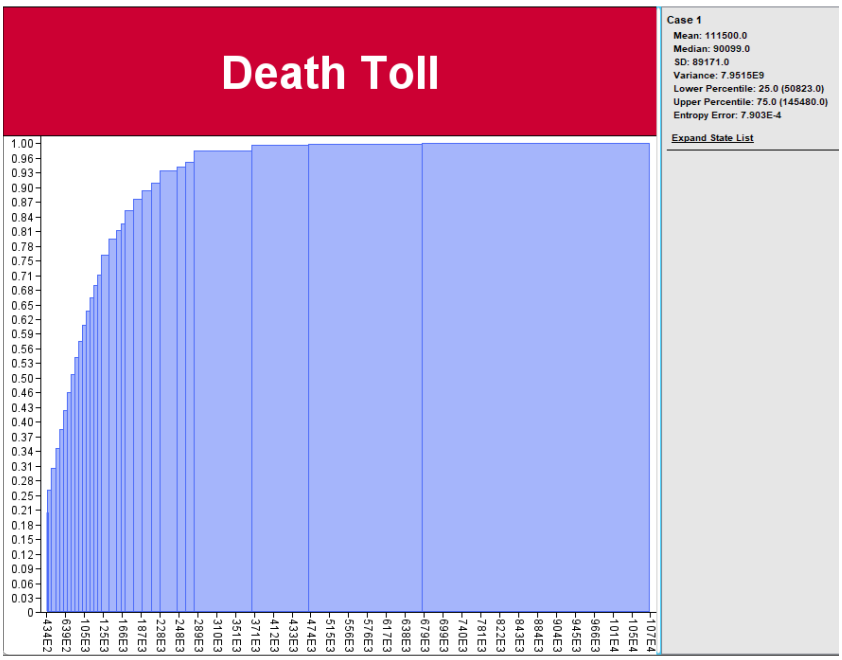


Figure B.9: Scenario 2, Question 2: Deaths in Italy

B.3. Scenario 3: Results

B.3.1. Question 1

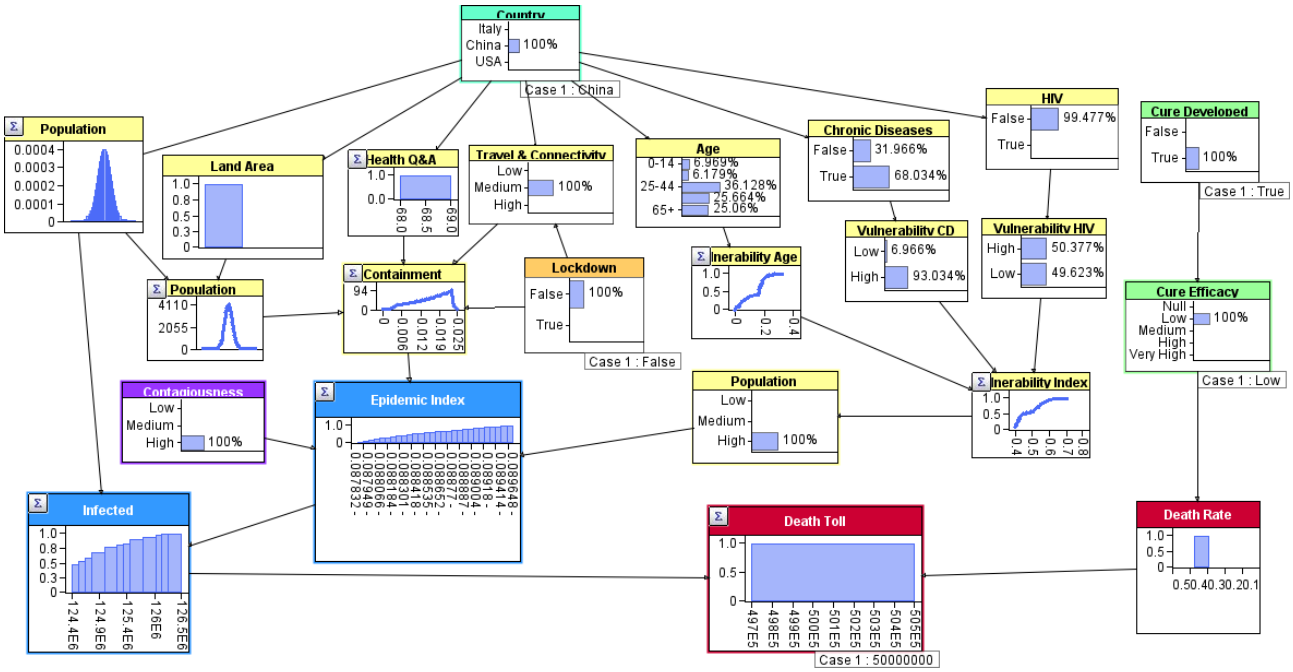


Figure B.10: Scenario 3, Question 1

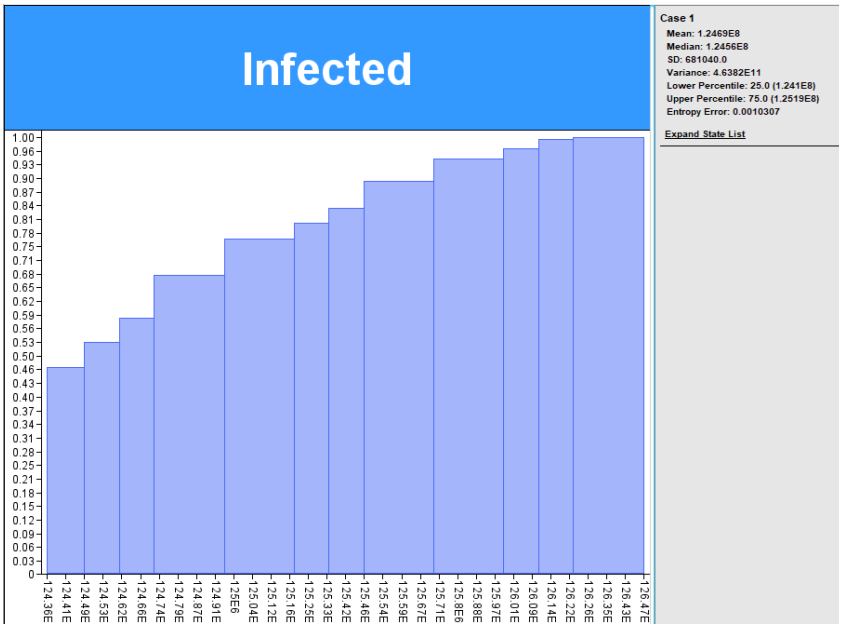
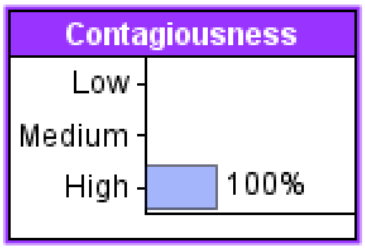
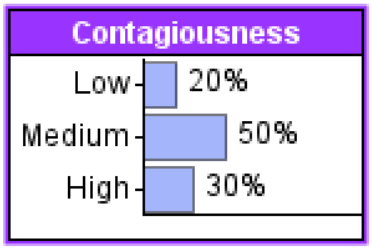


Figure B.11: Scenario 3, Question 1: Infected

B.3.2. Question 2



(a) Scenario's Consequences

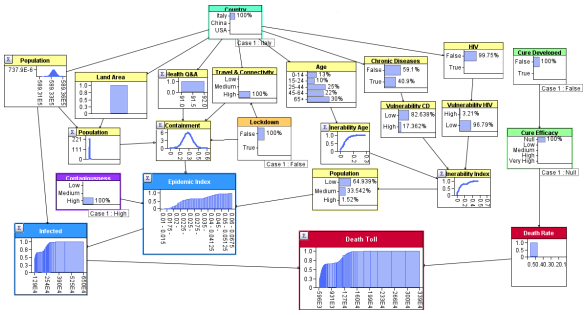


(b) Standard Contagiousness

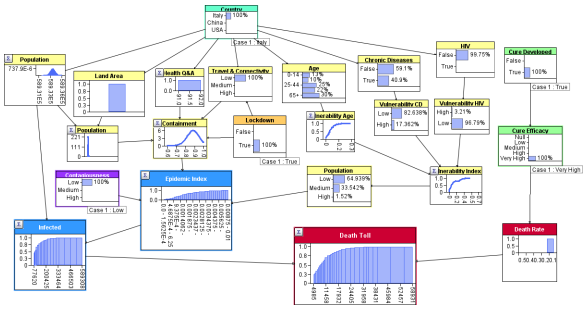
Figure B.12: Scenario 3, Question 2: Contagiousness Comparison

B.4. Scenario 4: Results

B.4.1. Question 1

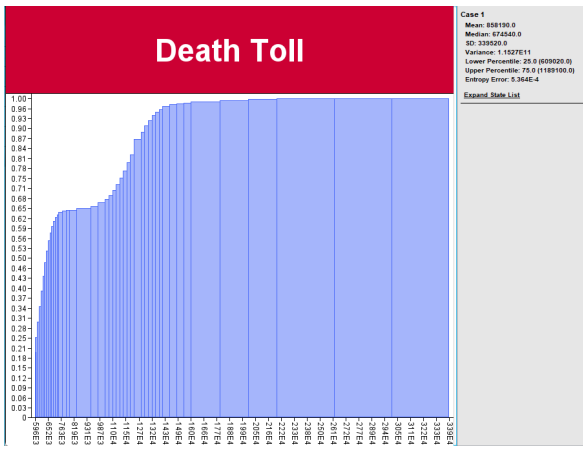


(a) Worst Case

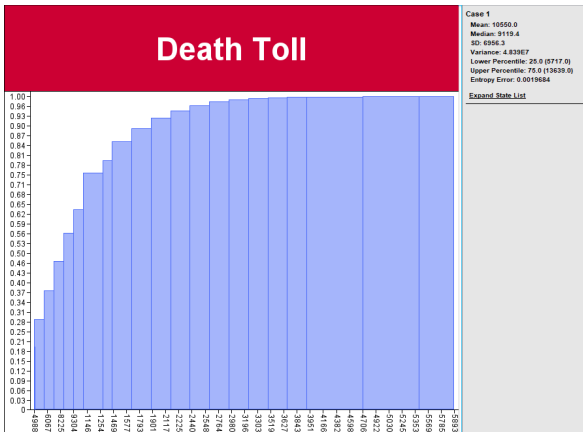


(b) Best Case

Figure B.13: Scenario 4



(a) Worst Case



(b) Best Case

Figure B.14: Scenario 4, Death Tolls

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